Height Growth Equations for CONUS: Douglas-fir

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2026-04-01

Data

We took the FIA height growth data subset for Douglas-fir, filtered out stray (likely keypunch errors) states, FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 141640 growth observations.

The observations are distributed among L3 Codes and states as follows:

NA_L3NAME	NA_L3CODE	n
Cascades	6.2.7	33303
Coast Range	7.1.8	23832
Klamath Mountains	6.2.11	16197
Columbia Mountains/Northern Rockies	6.2.3	15792
Middle Rockies	6.2.10	10533
Blue Mountains	6.2.9	7589
North Cascades	6.2.5	6849
Southern Rockies	6.2.14	5023
Idaho Batholith	6.2.15	4946
Eastern Cascades Slopes and Foothills	6.2.8	3582
Sierra Nevada	6.2.12	2674
Strait of Georgia/Puget Lowland	7.1.7	2620
Canadian Rockies	6.2.4	1968
Wasatch and Uinta Mountains	6.2.13	1328
Arizona/New Mexico Mountains	13.1.1	1271
Willamette Valley	7.1.9	1228
Colorado Plateaus	10.1.6	766
Northwestern Great Plains	9.3.3	679
Northern Basin and Range	10.1.3	491

NA_L3NAME	NA_L3CODE	n
California Coastal Sage, Chaparral, and Oak Woodlands	11.1.1	464
Columbia Plateau	10.1.2	284
Central Basin and Range	10.1.5	73
Madrean Archipelago	12.1.1	48
Northwestern Glaciated Plains	9.3.1	38
Wyoming Basin	10.1.4	35
Southwestern Tablelands	9.4.3	16
Arizona/New Mexico Plateau	10.1.7	10
Snake River Plain	10.1.8	1

State	n
OR	53782
WA	36005
MT	16919
CA	15378
ID	10557
CO	4503
UT	1827
NM	1161
AZ	908
WY	566
NV	34

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl - \beta_5 elev + \beta_6 * TD^{0.5} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 141640 growth observations, estimating β_0 through β_7. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- $elev$ = elevation above sea level (feet)
- TD = temperature difference between minimum and maximum ($^\circ C$)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for Douglas-fir. In our data that value is: 303 feet.

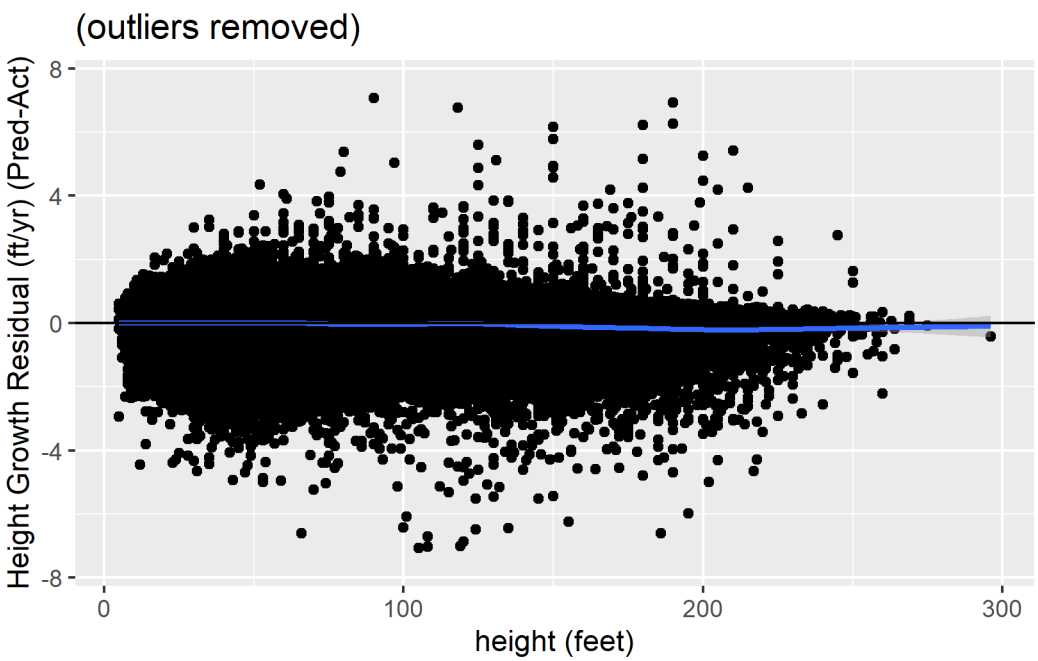
The parameter estimates and fit statistics are:

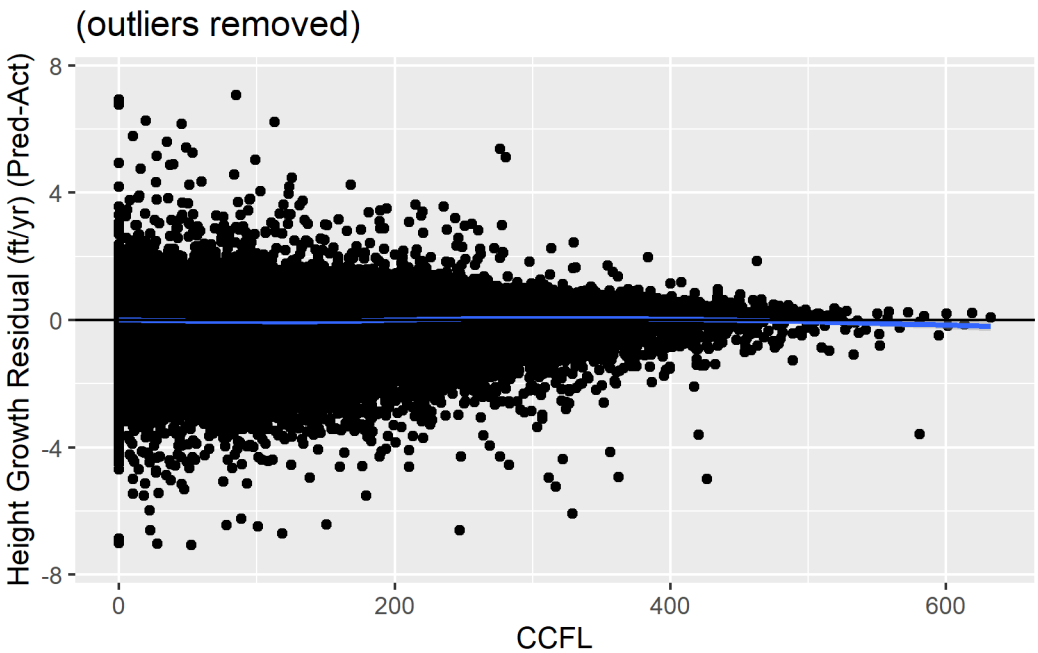
	Coef.	Std. error	t-stat.	p
b1	0.0108869	0.0001108	98.27453	0
b2	1.6261155	0.0137625	118.15517	0
b3	0.4573182	0.0065298	70.03581	0
b4	0.0014823	0.0000314	47.14924	0
b5	0.0001514	0.0000018	82.85264	0
b6	0.2736072	0.0026242	104.26418	0
b7	0.0326509	0.0004712	69.29141	0

```
Residual Standard Error: 6.72548726739847 on 141633 degrees of freedom. AIC =  941870.5
```

Residual Analysis for Equation 1

Mensurational Variables

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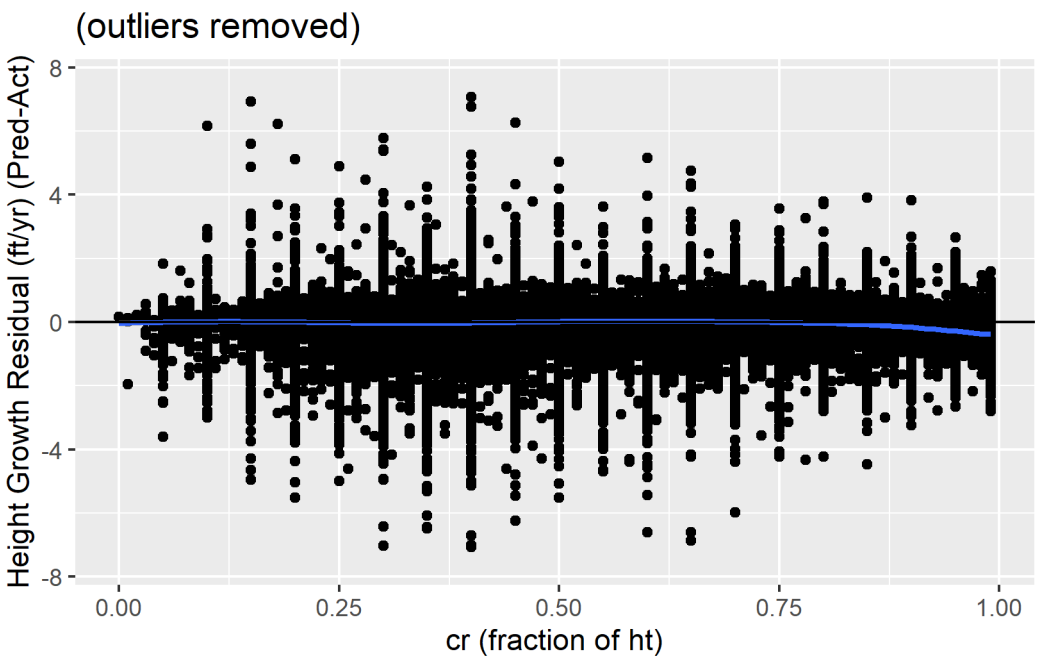
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
200
400
600
CCFL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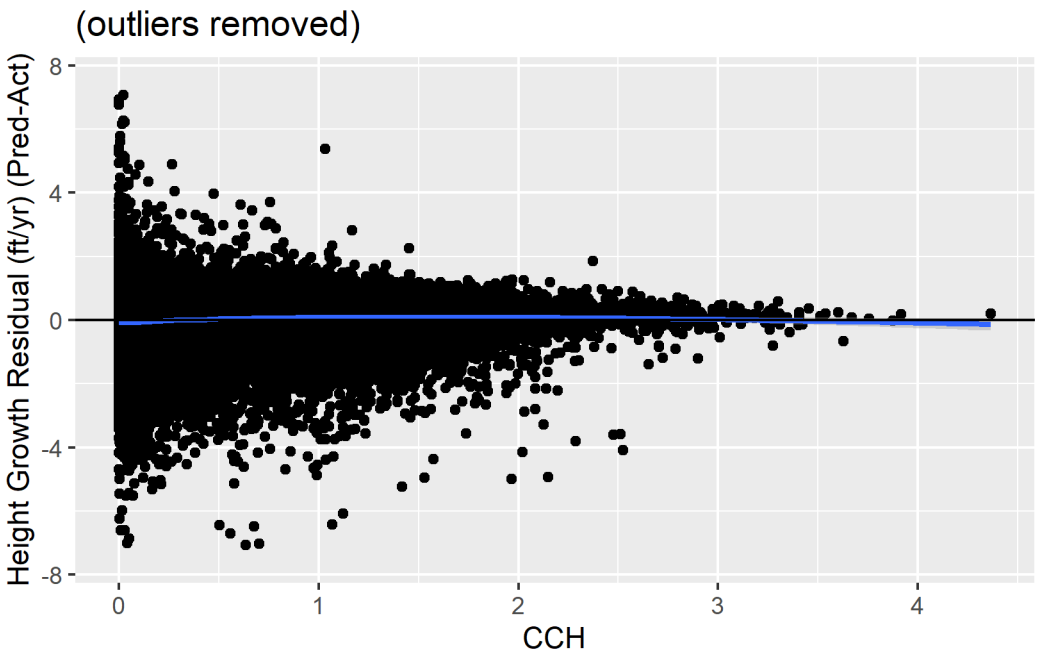
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)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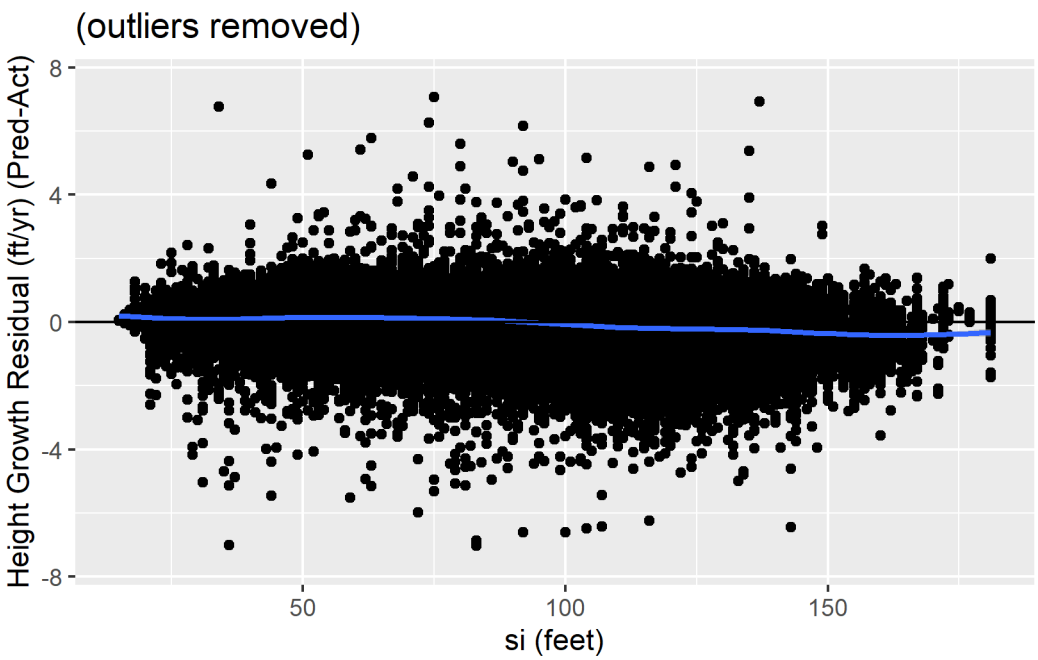
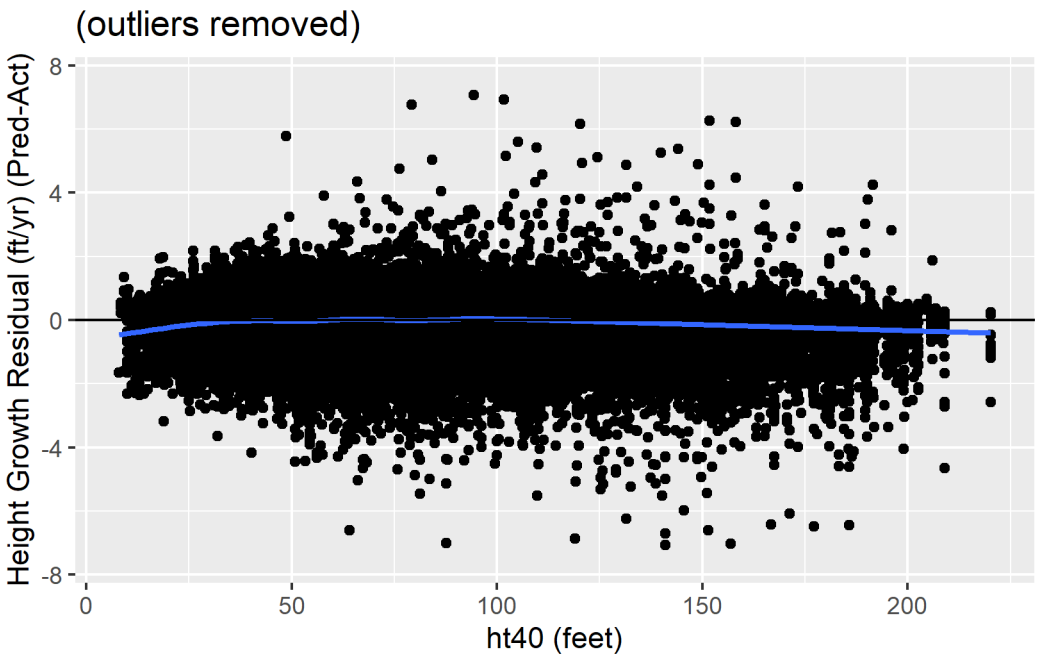
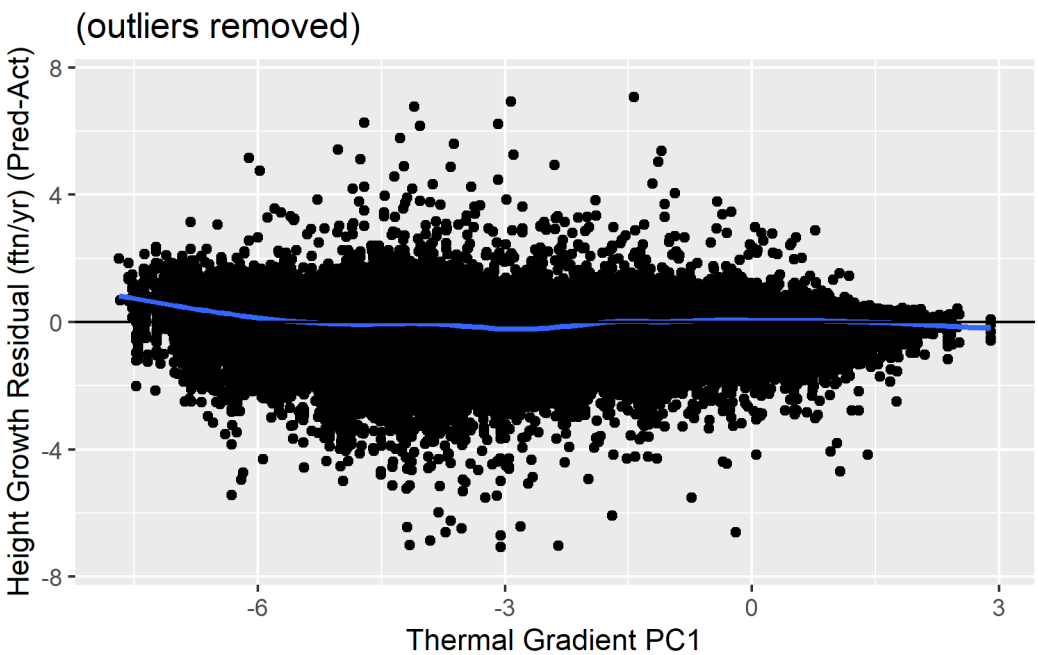
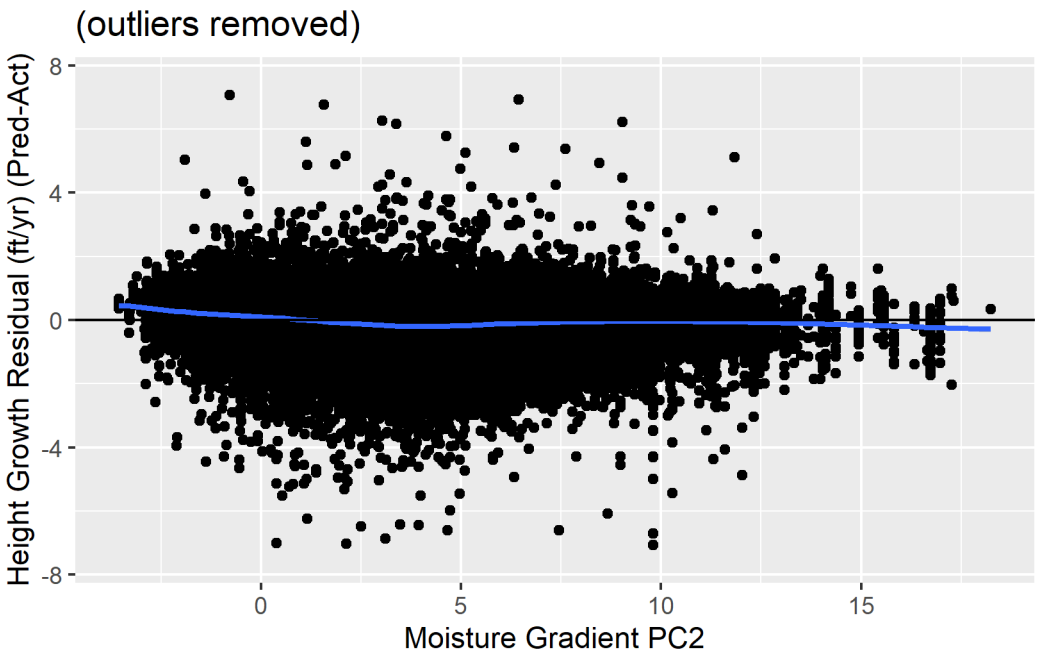
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
1
2
3
4
CCH


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
50
100
150
si (feet)



Environmental and Climate Variables

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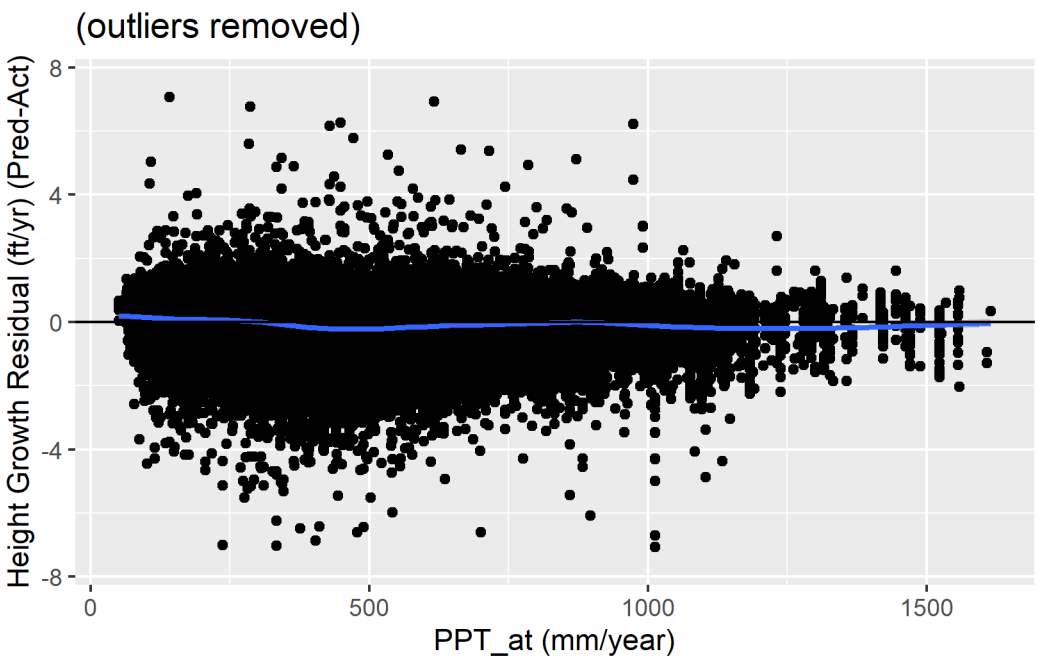
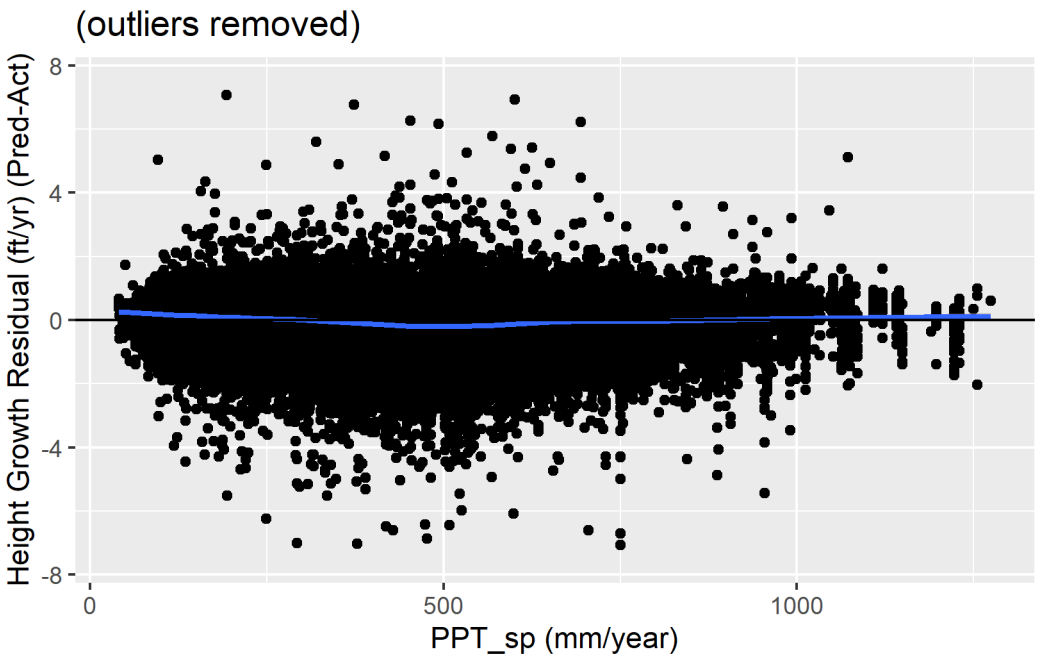
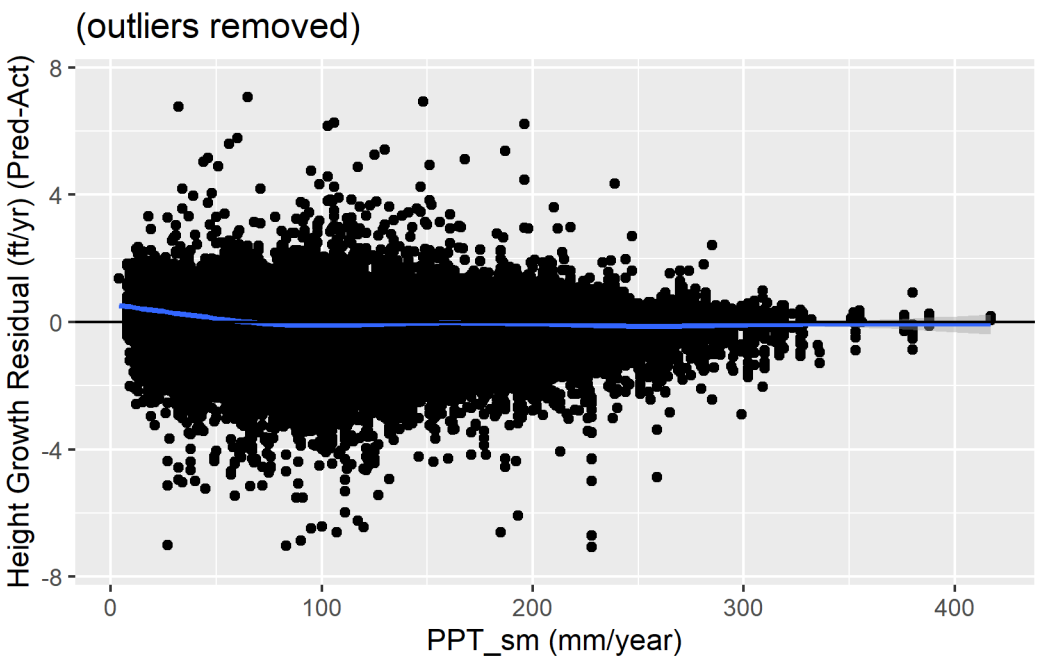
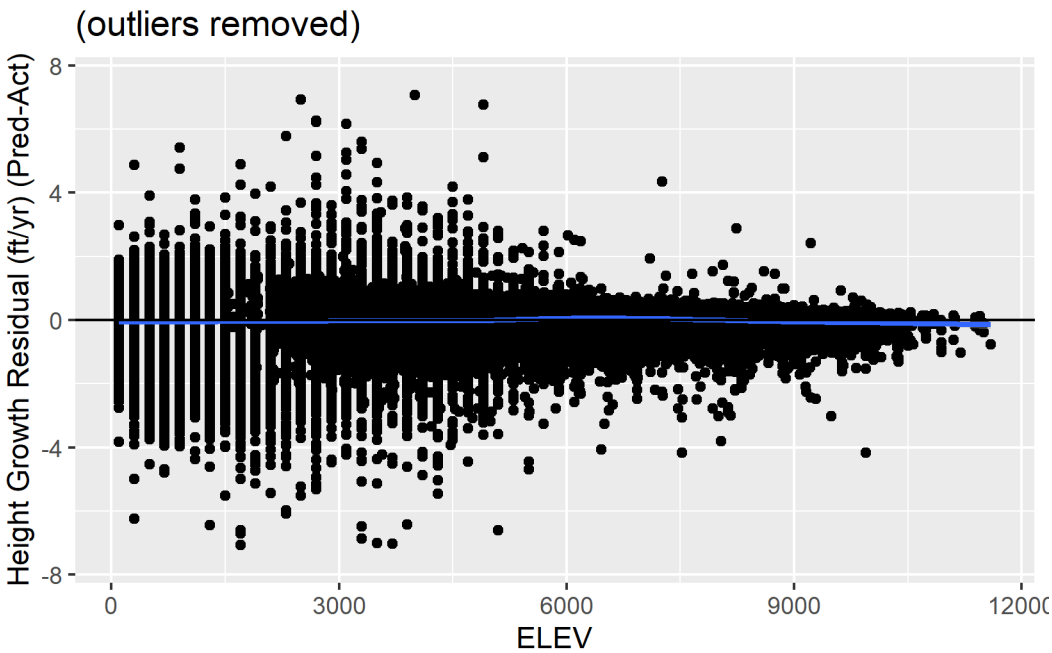
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
5
10
15
Moisture Gradient PC2


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
500
1000
1500
PPT_at (mm/year)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
500
1000
PPT_sp (mm/year)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
100
200
300
400
PPT_sm (mm/year)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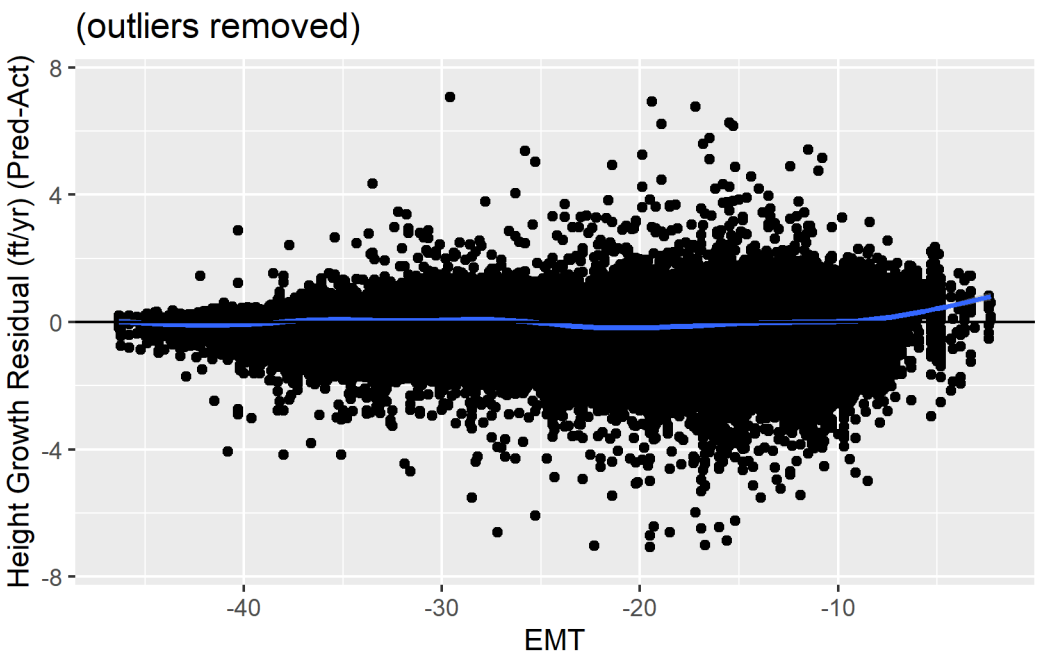
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
3000
6000
9000
12000
ELEV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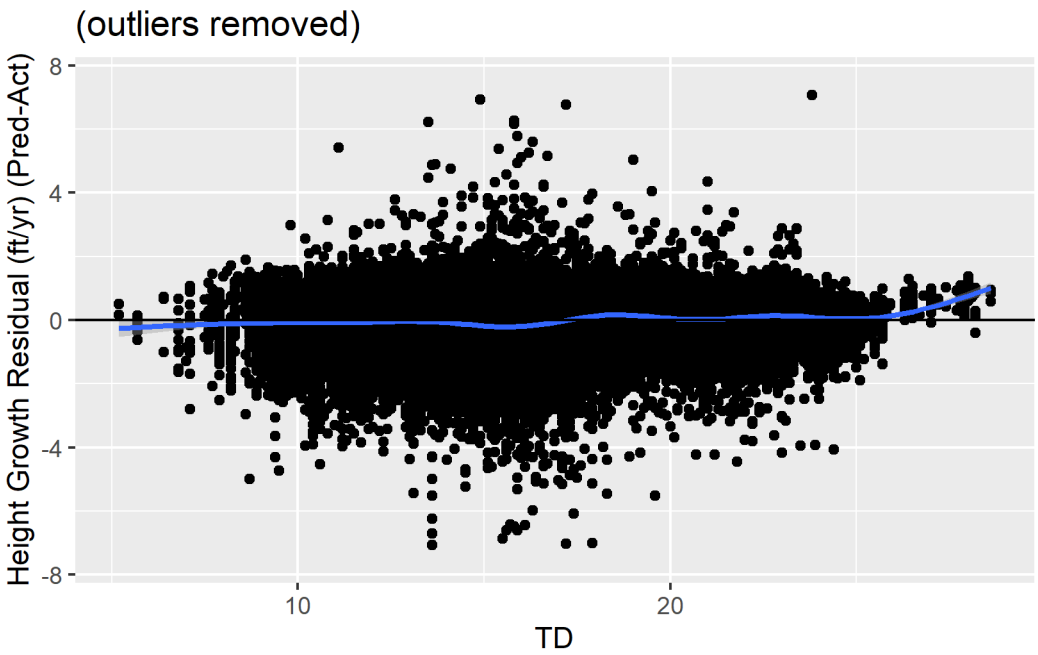
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
-40
-30
-20
-10
EMT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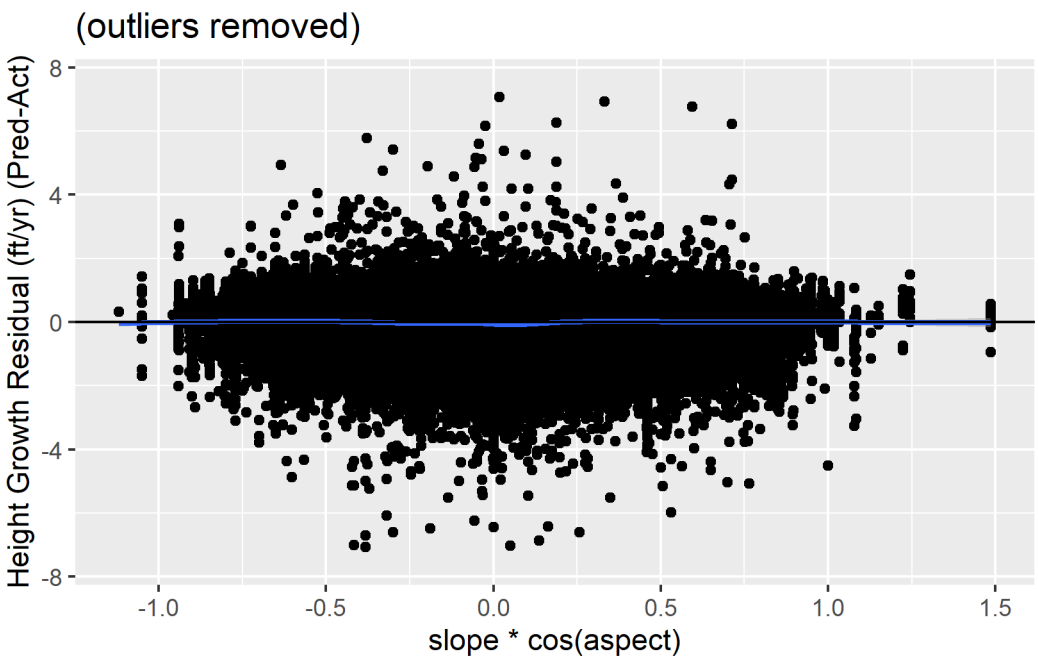
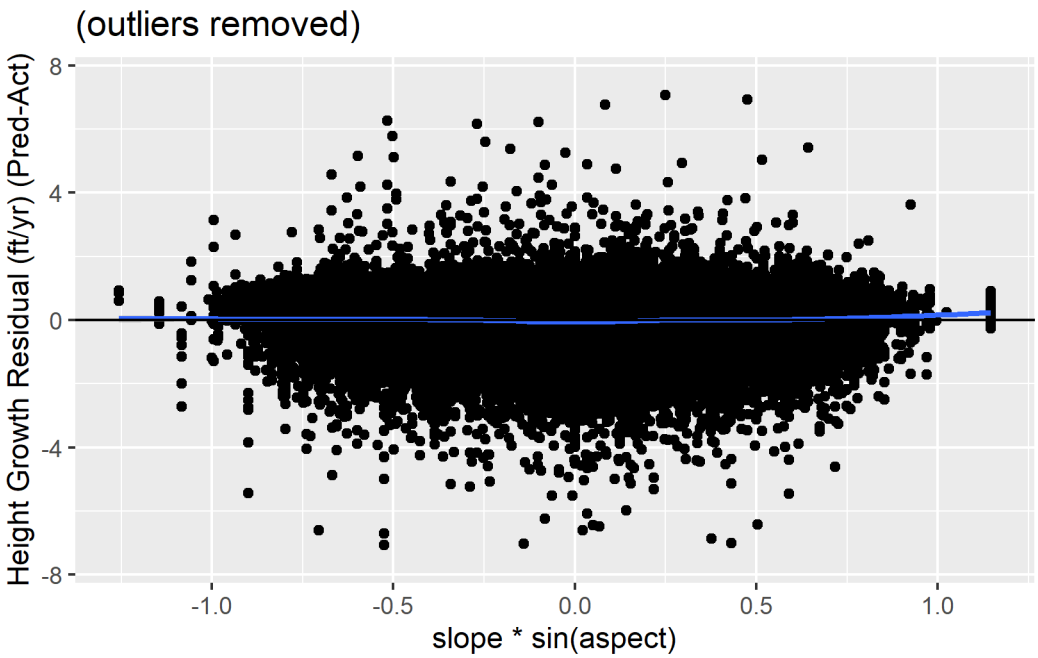
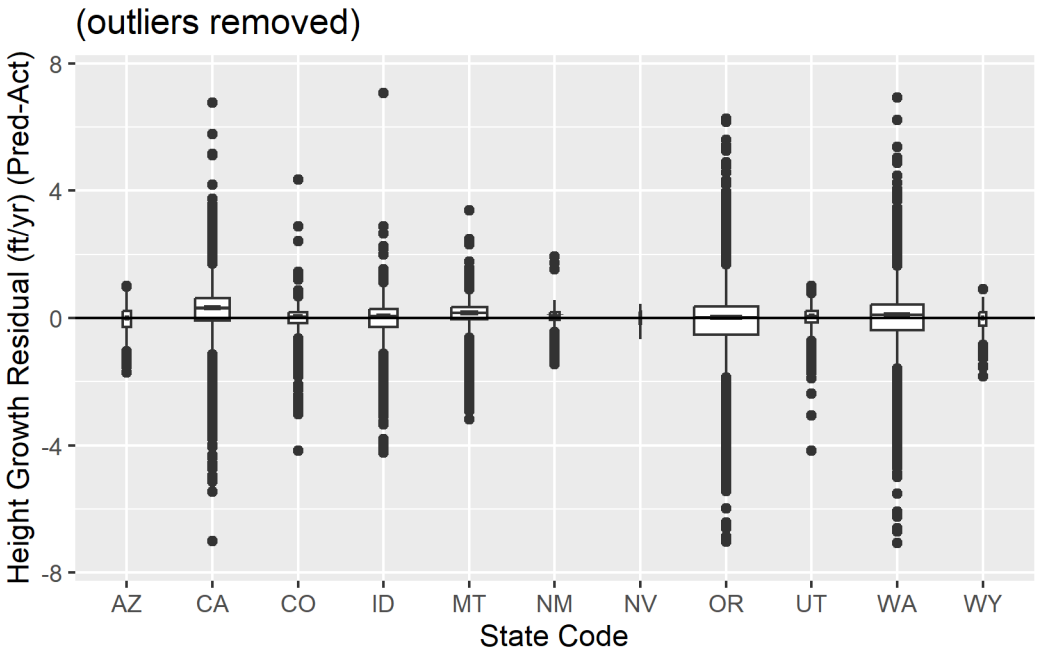
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
10
20
TD

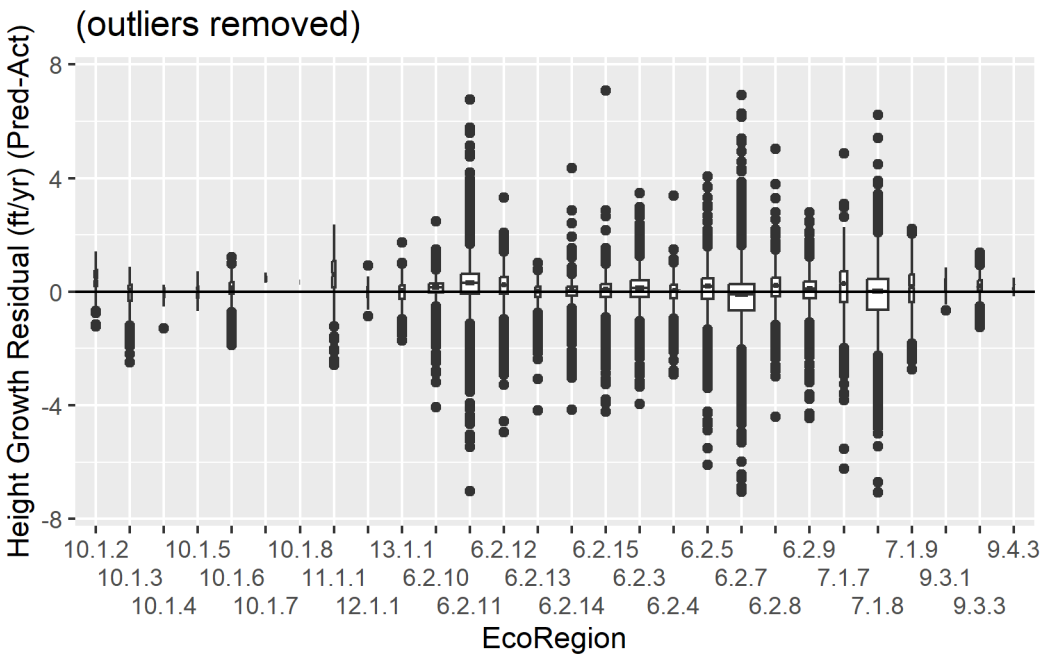

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
-1.0
-0.5
0.0
0.5
1.0
1.5
slope * cos(aspect)

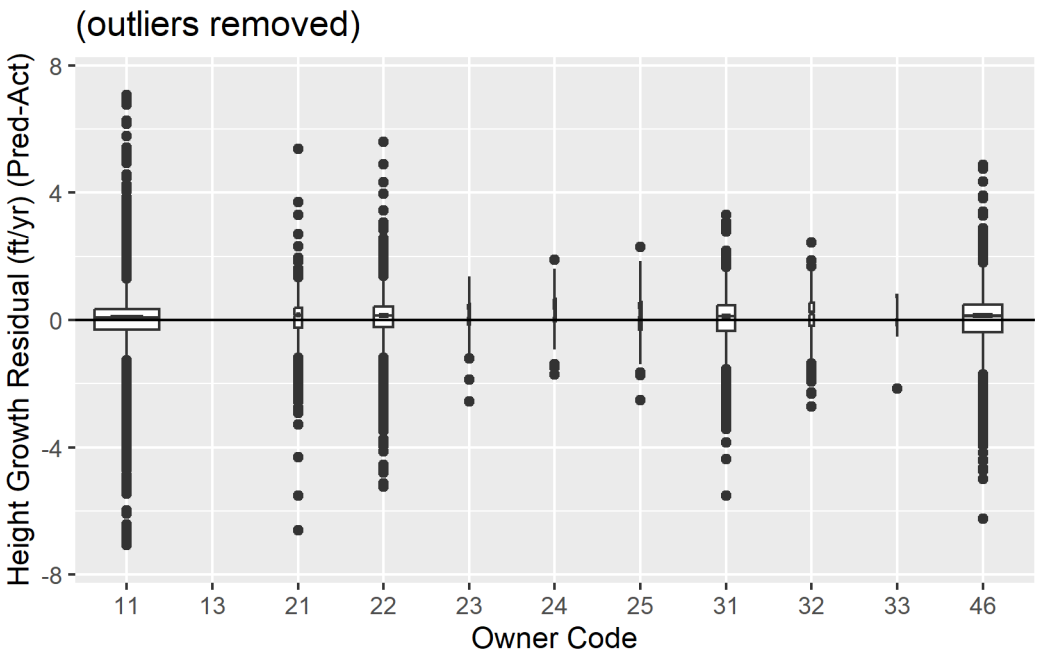


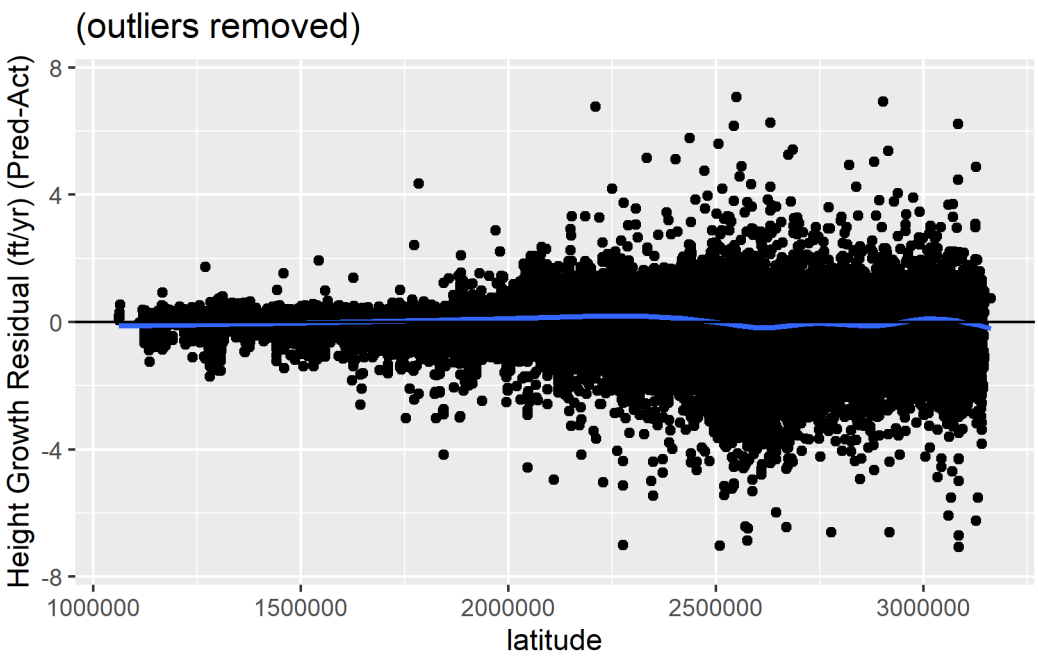

```
      hg_res          AHM         bFFP          CMD          CMI         DD_0
 1.000000000  0.165656091 -0.005848207  0.163647148 -0.129083905  0.052881825
       DD_18       DD1040         DD18          DD5         eFFP          EMT
-0.001212397  0.102348587  0.146747744  0.071045325 -0.022105723 -0.048580157
        Eref          EXT          FFP          MAP          MAT         MCMT
 0.071730076  0.099498109 -0.007151820 -0.110850423  0.017577000 -0.041891326
         MSP         MWMT         NFFD          PAS       PPT_at       PPT_sm
-0.147327316  0.126500858 -0.032105232 -0.045529695 -0.128665629 -0.093967126
      PPT_sp       PPT_wt           RH          SHM      Tave_at      Tave_sm
-0.121110681 -0.080972592 -0.101571710  0.186962774  0.010509421  0.120930838
     Tave_sp      Tave_wt           TD        lyr.1        lyr.2
 0.028531605 -0.039635784  0.124781420  0.016282039 -0.139385967
```

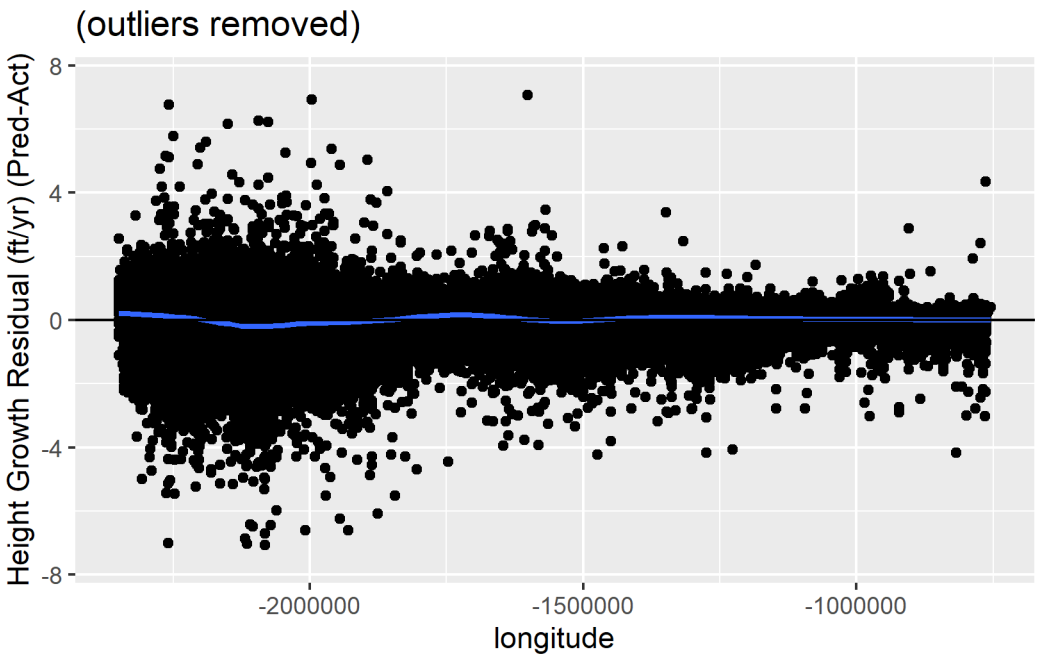
Location Variables

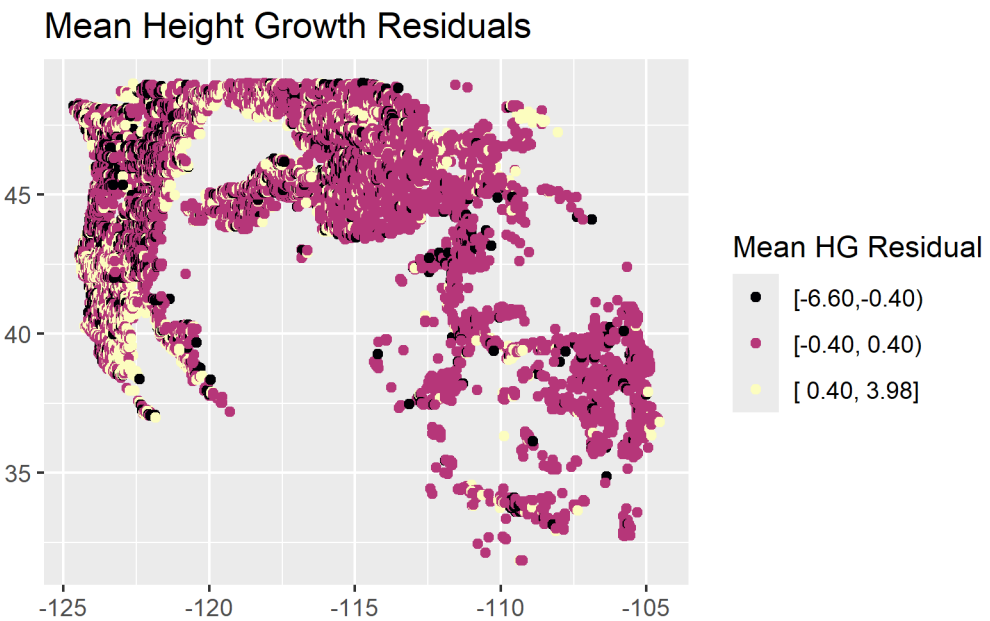




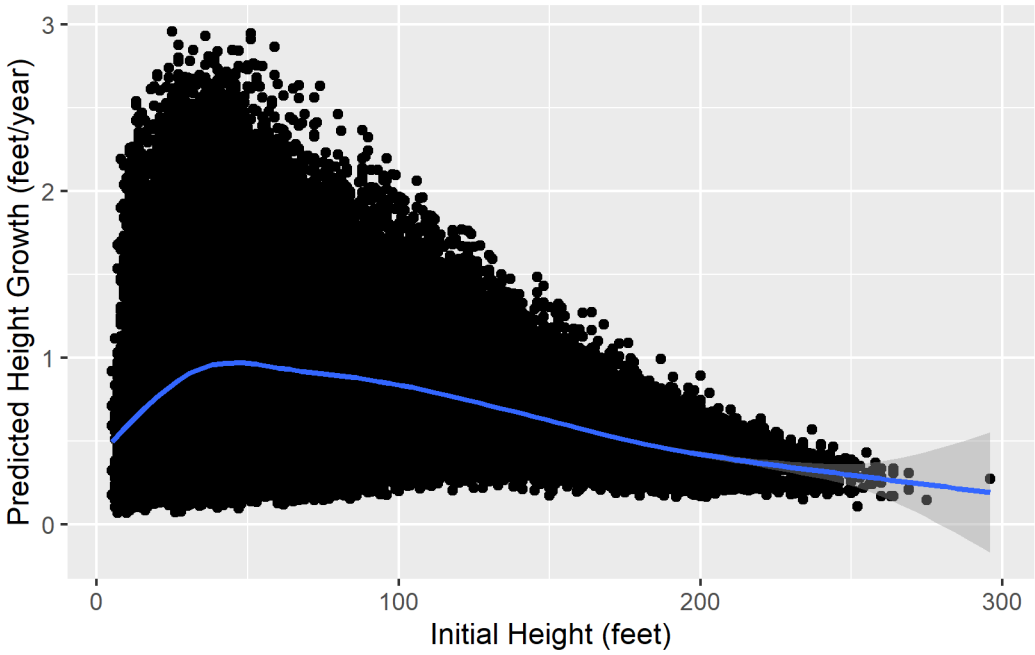

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
11
13
21
22
23
24
25
31
32
33
46
Owner Code

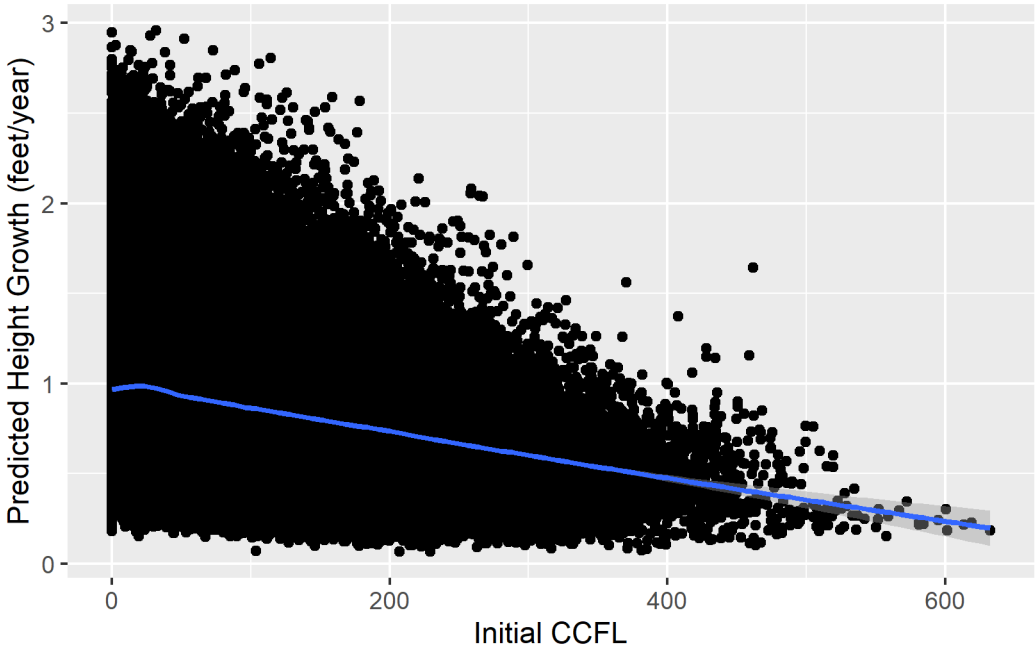

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
1000000
1500000
2000000
2500000
3000000
latitude

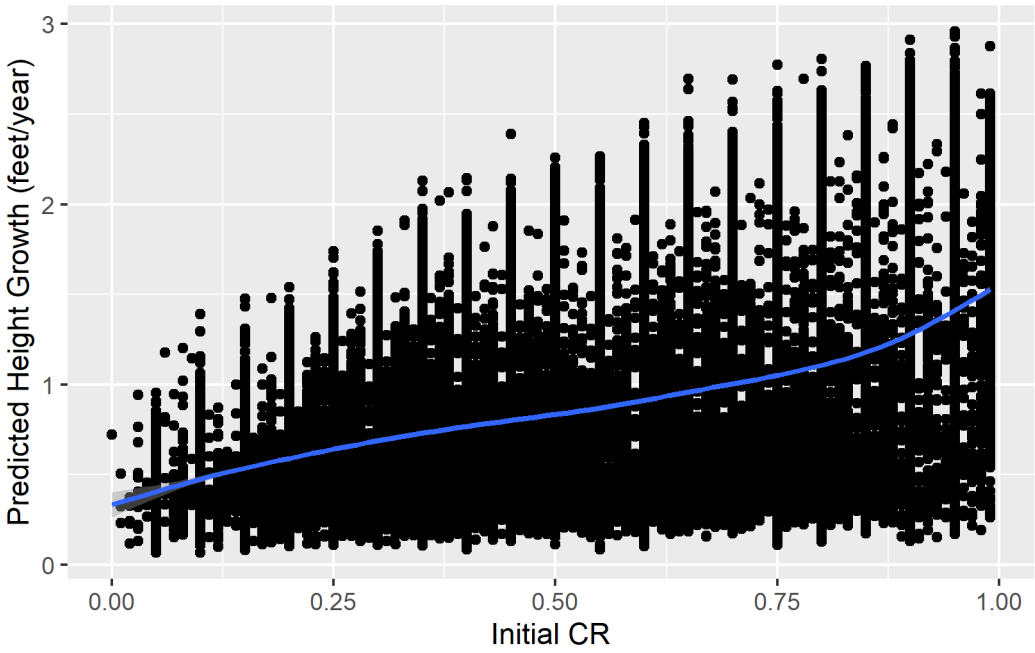

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
-2000000
-1500000
-1000000
longitude

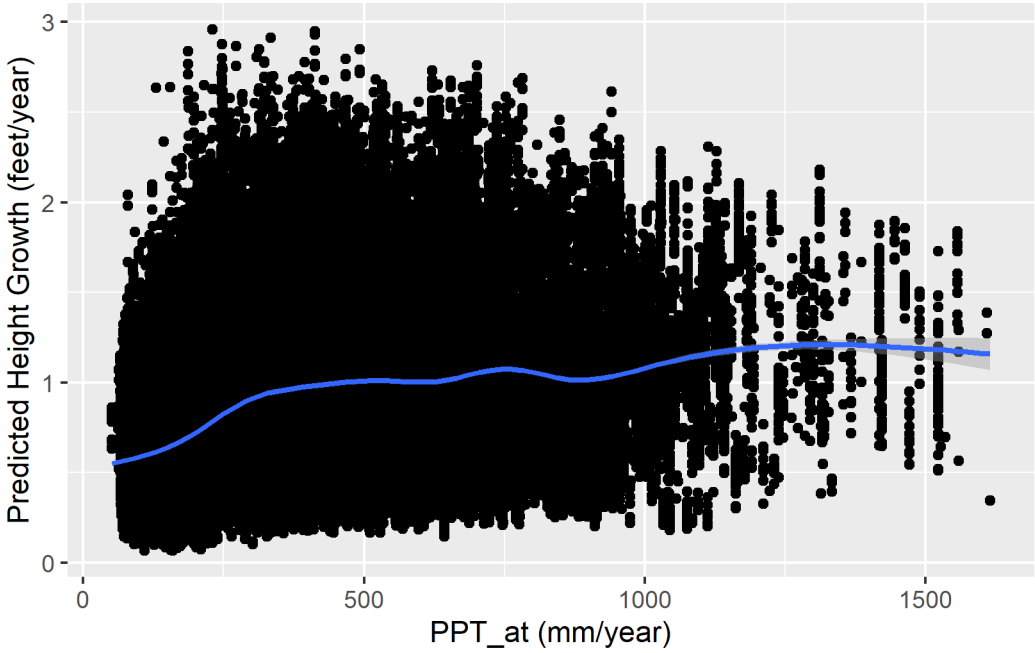

Mean Height Growth Residuals
45
40
35
-125
-120
-115
-110
-105
Mean HG Residual
[-6.60,-0.40)
[-0.40, 0.40)
[0.40, 3.98]

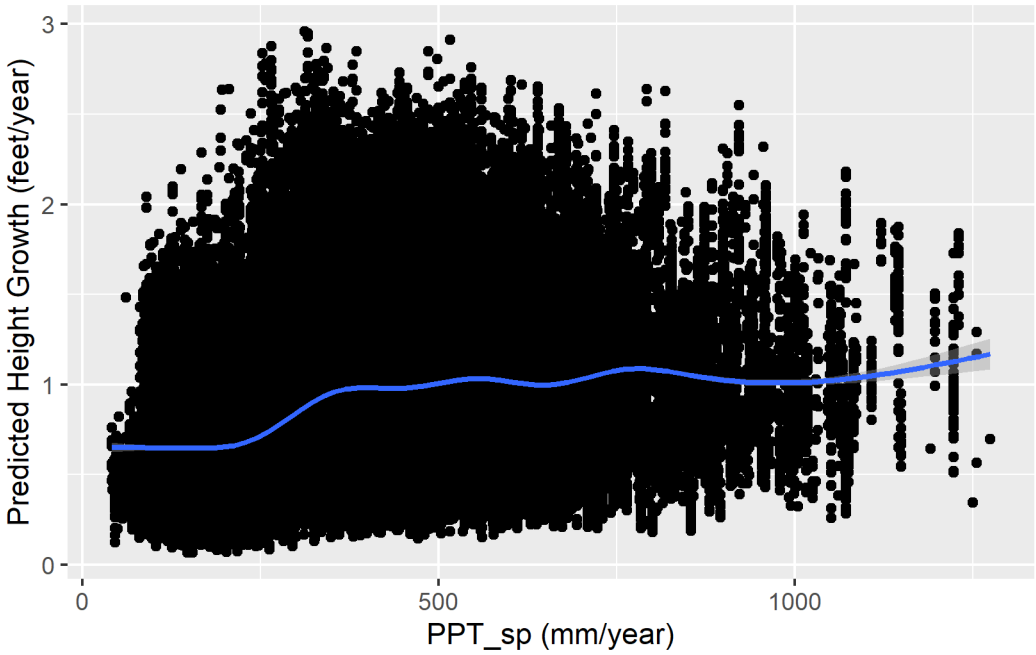
Prediction Performance

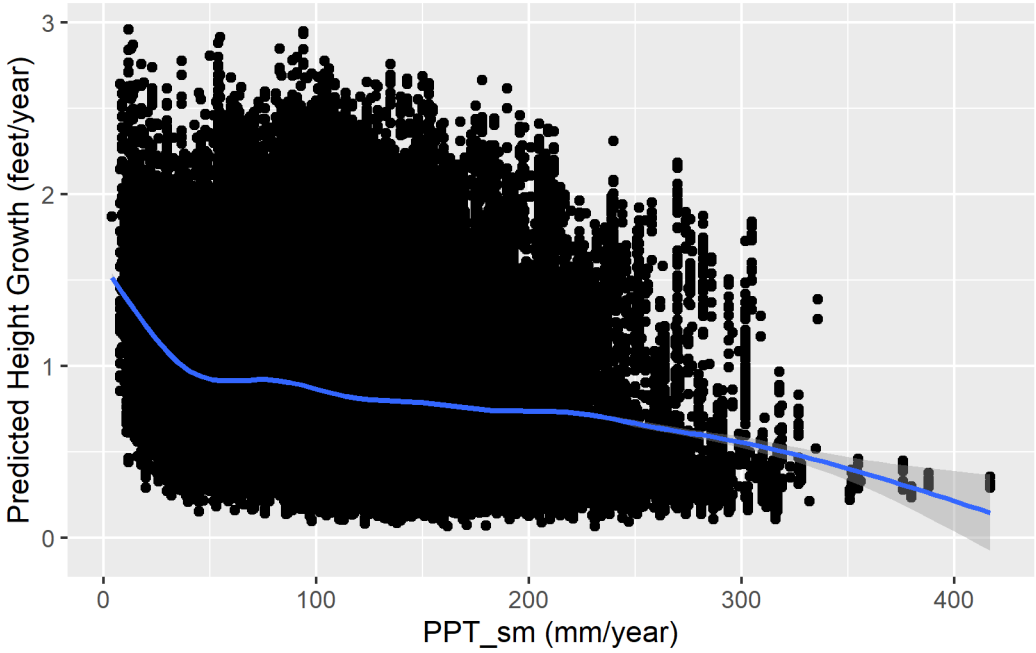


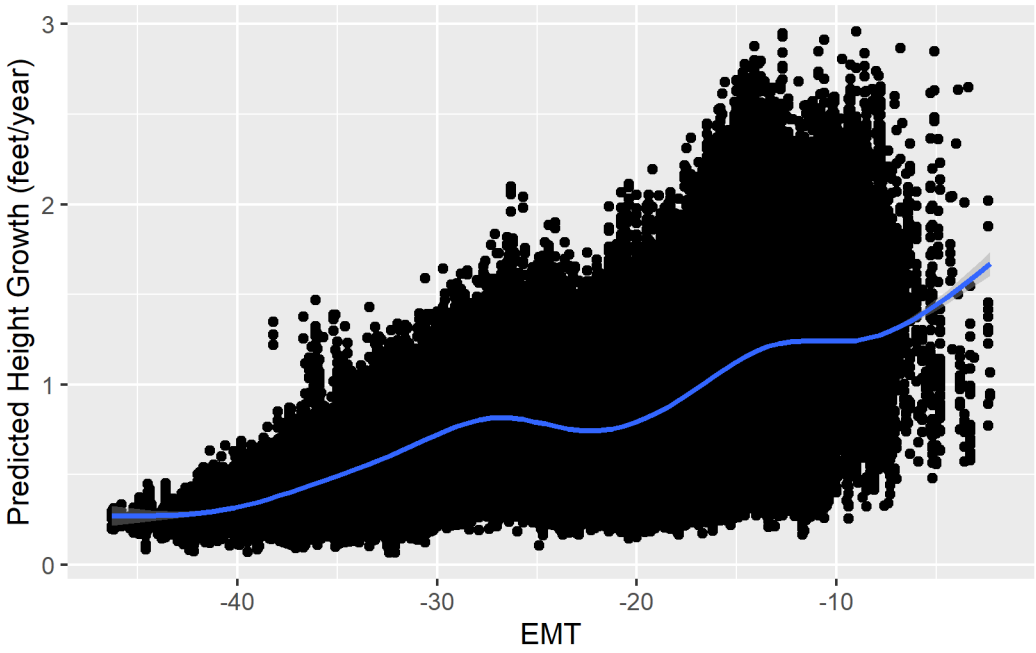



Predicted Height Growth (feet/year)
3
2
1
0
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
3
2
1
0
0
500
1000
1500
PPT_at (mm/year)


Predicted Height Growth (feet/year)
3
2
1
0
0
500
1000
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
3
2
1
0
0
100
200
300
400
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
3
2
1
0
-40
-30
-20
-10
EMT